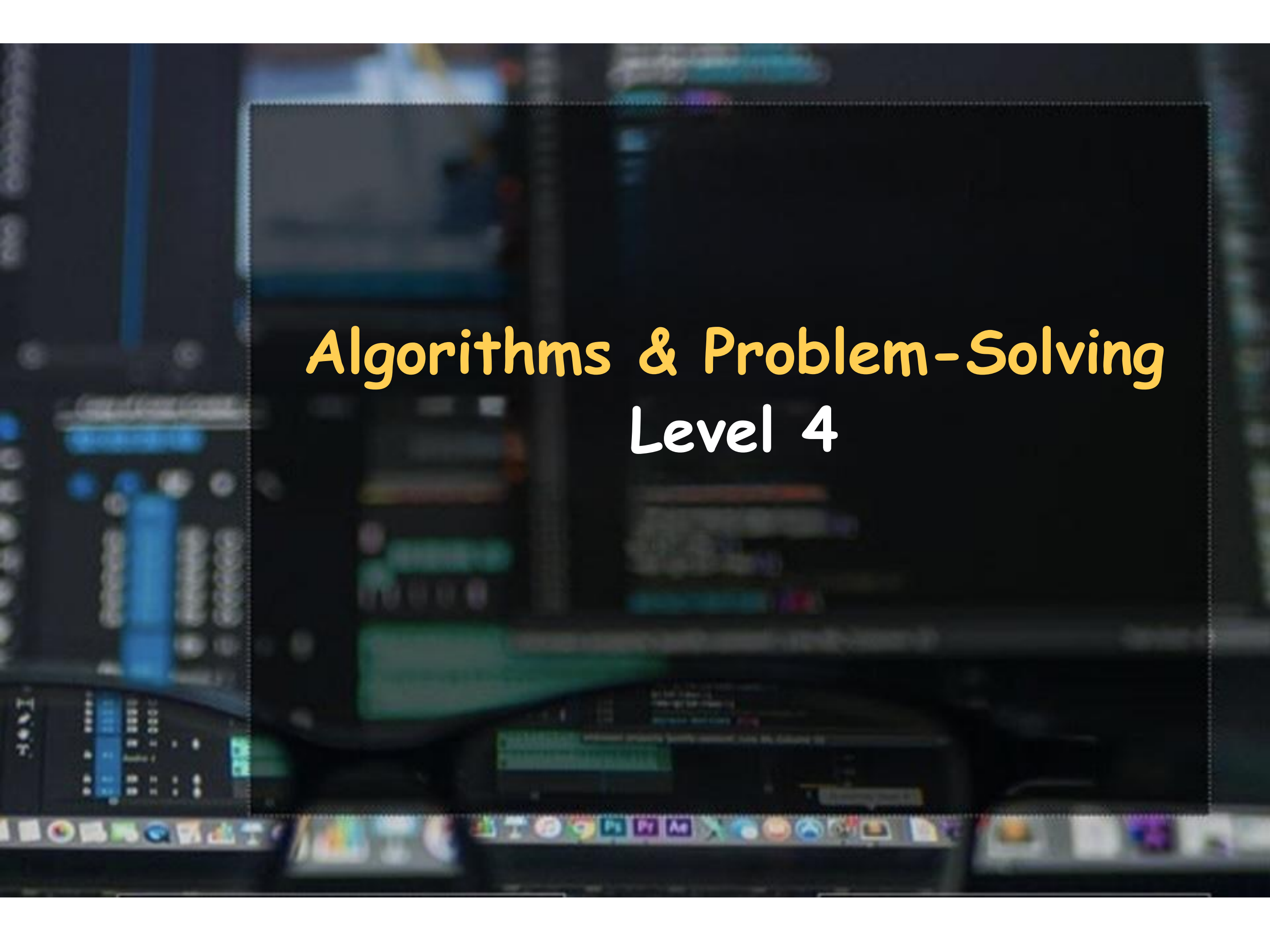




Algorithms & Problem-Solving Level 4



Algorithms & Problem Solving

Level 4

Diff In Days (Negative Days)



Definition

- The goal is a program to calculate the difference between two dates (in days), taking into account the possibility that the difference will be negative if the second date is earlier than the first.



Explanation

- The goal is a program that reads two dates from the user (day, month, year) and calculates the difference between them in days.
- If the second date is older, the result is printed as negative.
- The difference can also be calculated with or without including the last day.

Procedures

- 1 Defining the structure:
 -  The sDate structure contains:
 -  Day, Month, Year
- 2 Data input:
 -  Functions to read the day, month, and year and group them into a variable of type sDate
- 3 Leap year checking:
 -  IsLeapYear() checks if the year is a leap year to adjust the number of days in February
- 4 Comparing dates:
 -  IsDate1BeforeDate2() determine which is older
- 5 Number of days in a month:
 -  NumberOfDaysInAMonth() returns the number of days, taking into account February and leap years
- 6 Increasing the day:
 -  IncreaseDateByOneDay() increments the date by one day, jumping to the next month or year if needed
- 7 Swapping dates:
 -  SwapDates() if the first date is newer, swaps it so that the first date is the newer one Oldest.

8 Calculating the difference:

- `GetDifferenceInDays()` calculates the number of days between the two dates.
- If `includeEndDay` is true, the end day is added to the difference.



Example:

Date1 = 1/1/2000

Date2 = 1/1/2022

Difference is: 8036 Day(s)

Difference (Including End Day): 8037 Day(s)

Date1 = 1/1/2022

Date2 = 1/1/2000

Difference is: -8036 Day(s)

Difference (Including End Day): -8037 Day(s)



Note:

- The program automatically returns the negative difference when reversing dates thanks to the `SwapDates()` function.
- The calculation is done by repeatedly incrementing the date until the second date is reached.

Summary

- ✓ The program links time calculations with logical date comparisons.
- ✓ It explains the concept of iteration by manually adding days.
- ✓ It supports negative difference and flexible inclusion of the end date.

Thank you 😊